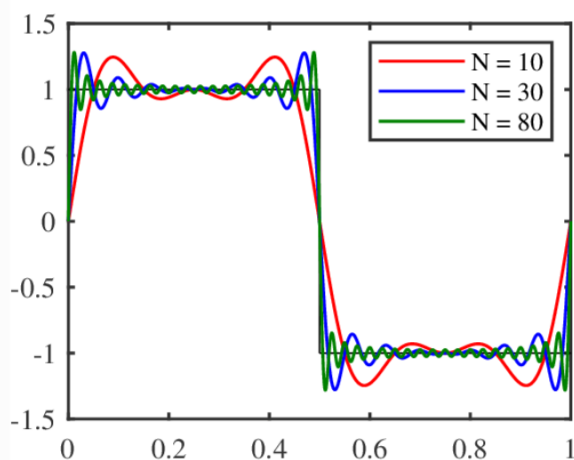
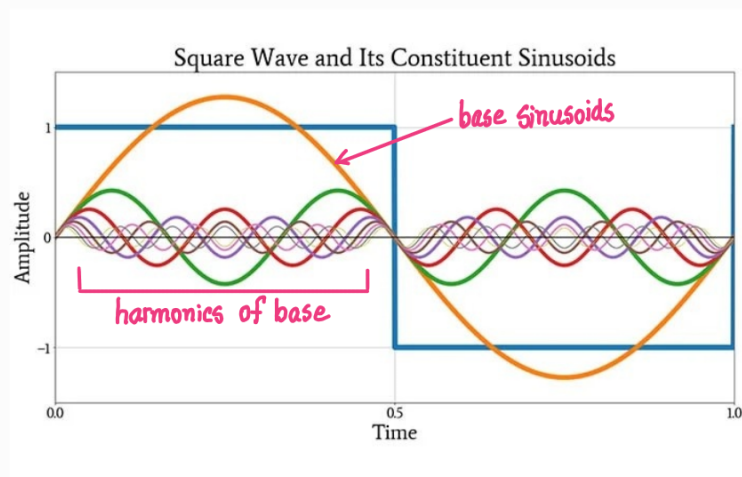


Fourier Series —

We will start our journey into the frequency domain with Fourier series. We have most concepts that we need. If we need anything more we will explore midway.

Fourier states that any periodic signal can be broken into a weighted sum of sinusoids (combination of sine and cosine). But to maintain periodicity, all the sinusoids must have same frequency as the original signal. The frequency doesn't have to be fundamental frequency obvio.



The base sinusoid will have same fundamental frequency (f_0) as the original signal. All other constituent signals will have frequencies that are integer multiples of f . These signals having kf_0 frequency where $k \in \mathbb{Z}$ are called harmonics of the base sinusoid.

$$x(t) = \sum_{k=-\infty}^{\infty} a_k \cos 2\pi k f_0 t + \sum_{k=-\infty}^{\infty} b_k \sin 2\pi k f_0 t \quad \text{Synthesis Equation}$$

Here, a_k denotes how much the cosine term with $f = kf_0$ contributes to the summation on average and b_k does the same for the sine terms. To find the value of a_k we simply find the full overlap correlation rate between the cosine ($f = kf_0$) and one oscillation of the signal —

$$a_k = \frac{1}{T} \int_0^T x(t) \cdot \cos 2\pi k f_0 t \, dt \quad \text{Analysis equation 1}$$

Similarly, $b_k = \frac{1}{T} \int_0^T x(t) \cdot \sin 2\pi k f_0 t \, dt$ *Analysis equation 2*

We can update the synthesis equation a bit. But first we need to derive a few things —

$$a_0 = \frac{1}{T} \int_0^T x(t) \cdot \cos 0^\circ \, dt = \frac{1}{T} \int_0^T x(t) \, dt \quad \text{--- (i)}$$

$$a_{-k} = \frac{1}{T} \int_0^T x(t) \cdot \cos(-2\pi k f_0 t) \, dt = \frac{1}{T} \int_0^T x(t) \cdot \cos(2\pi k f_0 t) \, dt \quad \left[\cos x \text{ is an even signal} \right. \\ \left. \rightarrow \cos(-x) = \cos(x) \right]$$

$$\therefore a_{-k} = a_k \quad \text{--- (ii)}$$

$$b_{-k} = \frac{1}{T} \int_0^T x(t) \cdot \sin(-2\pi k f_0 t) \, dt = -\frac{1}{T} \int_0^T x(t) \cdot \sin 2\pi k f_0 t \, dt \quad \left[\sin x \text{ is an odd function} \right. \\ \left. \rightarrow \sin(-x) = -\sin(x) \right]$$

$$\therefore b_{-k} = -b_k \quad \text{--- (iii)}$$

Similarly you can prove very easily that $b_0 = 0$ --- (iv)

Now let's update the original synthesis function a bit —

$$x(t) = \sum_{k=-\infty}^{\infty} a_k \cos 2\pi k f_0 t + \sum_{k=-\infty}^{\infty} b_k \sin 2\pi k f_0 t$$

$$= \left(\sum_{k=-\infty}^{\infty} a_k \cos 2\pi k f_0 t \right) + \left(\sum_{k=-\infty}^{\infty} b_k \sin 2\pi k f_0 t \right)$$

$$= \left(\sum_{k=-\infty}^{-1} a_k \cos 2\pi k f_0 t + \underbrace{a_0 \cos 0^\circ}_{a_0} + \sum_{k=1}^{\infty} a_k \cos 2\pi k f_0 t \right)$$

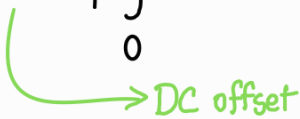
replacing k by -k

$$+ \left(\sum_{k=-\infty}^{-1} b_k \sin 2\pi k f_0 t + \underbrace{b_0 \sin 0^\circ}_0 + \sum_{k=1}^{\infty} b_k \sin 2\pi k f_0 t \right)$$

replacing k by -k

$$\begin{aligned}
&= a_0 + \left(\sum_{k=1}^{\infty} a_{-k} \cos(-2\pi k f_0 t) + \sum_{k=1}^{\infty} a_k \cos 2\pi k f_0 t \right) \\
&\quad + \left(\sum_{k=1}^{\infty} b_{-k} \sin(-2\pi k f_0 t) + \sum_{k=1}^{\infty} b_k \sin 2\pi k f_0 t \right) \\
&= a_0 + \left(\sum_{k=1}^{\infty} a_k \cos 2\pi k f_0 t + \sum_{k=1}^{\infty} a_k \cos 2\pi k f_0 t \right) \\
&\quad + \left(\sum_{k=1}^{\infty} (-b_k) (-\sin 2\pi k f_0 t) + \sum_{k=1}^{\infty} b_k \sin 2\pi k f_0 t \right) \\
&= a_0 + 2 \sum_{k=1}^{\infty} a_k \cos 2\pi k f_0 t + 2 \sum_{k=1}^{\infty} b_k \sin 2\pi k f_0 t \\
&= a_0 + \sum_{k=1}^{\infty} a'_k \cos 2\pi k f_0 t + \sum_{k=1}^{\infty} b'_k \sin 2\pi k f_0 t
\end{aligned}$$

Here, $a_0 = \frac{1}{T} \int_0^T x(t) dt$; $a'_k = \frac{2}{T} \int_0^T x(t) \cdot \cos 2\pi k f_0 t dt$; $b'_k = \frac{2}{T} \int_0^T x(t) \cdot \sin 2\pi k f_0 t dt$

 DC offset

This is most probably the version you studied in your math course.

Questions so far —

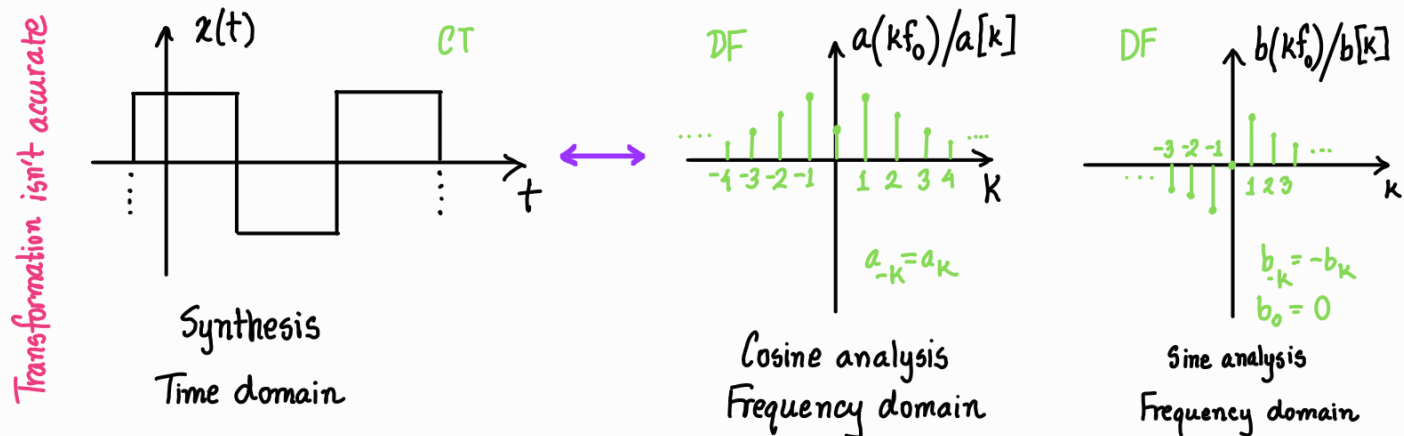
Question — Why do we not include the middle frequencies, that are not harmonics of the base signal in the summation?

Answer — It would add unequal contribution to different cycles causing inconsistencies.

Question — What are the synthesis and analysis equations?

Answer — The analysis equations give you the sine and cosine contribution for each frequency value and is the frequency domain representation. The synthesis

Equation gathers all the frequency contributions and return the time representation again.



Question — Why did we use full overlap correlation rate?

Answer — The sliding window correlation accounts phase shift. Since we have already divided the phase shift into sine and cosine (concept of orthogonal basis), we don't need to use the sliding window. The full overlap maintains orthogonality between the sine and the cosine terms. Taking the rate of correlation gives us the average correlation value.

Question — Which form is better? the first one with k varying from $-\infty$ to $+\infty$? Or the second with k varying from 0 to $+\infty$?

Answer — From here onwards we will work with frequencies ranging from $-\infty$ to $+\infty$. Although negative frequencies won't make much sense with real signals, it will help us a lot mathematically. You can always adjust your frequency domain by discarding the -ve frequencies and doubling the positive frequencies except the DC values for real analysis.

The current trigonometric form is cool but there are issues that we need to address.

Shortcomings of trigonometric Fourier series —

First of all, the sine and cosine decomposition is brilliant, but it gives you two different frequency domain graphs. This is something that we don't like. We just want to know the contribution of a certain frequency without breaking it into sine and cosine. Additionally, there should be only a sine or a cosine signal shifted by some phase instead of having both. But our trigonometric Fourier series can't do either directly. So it kinda sucks from this angle, let alone the fact that you are doing double the calculation, once for cosine and once for sine. And personally I hate working with trigonometric integrals as they require pattern matching at the end which is a struggle. So let's build towards something else.

We will start by figuring out the magnitude and the phase from trigonometric Fourier series —

In the synthesis function defined for $k = -\infty$ to $+\infty$, for any k value we get —

$$x_k(t) = a_k \cos \omega_k t + b_k \sin \omega_k t$$

$$\text{Let } x_k(t) = A_k \cos(\omega_k t - \phi) = A_k \cos \phi \cos \omega_k t + A_k \sin \phi \sin \omega_k t \quad [\text{formula for } \cos(a-b)]$$

$$\text{Comparing we get, } A_k \cos \phi = a_k \text{ \& } A_k \sin \phi = b_k$$

$$\text{Then, } A_k^2 \cos^2 \phi = a_k^2 \text{ \& } A_k^2 \sin^2 \phi = b_k^2$$

$$\text{Implies, } A_k^2 \cos^2 \phi + A_k^2 \sin^2 \phi = a_k^2 + b_k^2$$

$$\Rightarrow A_k^2 = a_k^2 + b_k^2 \Rightarrow A_k = \sqrt{a_k^2 + b_k^2}$$

$$\text{Also, } \frac{A_k \sin \phi}{A_k \cos \phi} = \frac{b_k}{a_k}$$

$$\Rightarrow \tan \phi = \frac{b_k}{a_k} \Rightarrow \phi = \tan^{-1} \frac{b_k}{a_k}$$

If you haven't noticed where we are going with this, we are going for complex sinusoid implementation of Fourier series. It will make things more compact and will cut computation by half.

Using complex exponentials simplifies the synthesis equation —

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} \longrightarrow \text{doesn't contain phase part just like trig variant}$$

$\rightarrow$ I know that it seems like we are ignoring the phase part, but bear with me for a sec

Now all we have to do is figure out the analysis equation

multiplying both sides of the synthesis by $e^{-jn\omega_0 t}$ where $n \in \mathbb{Z}$ —

$$x(t) \cdot e^{-jn\omega_0 t} = e^{-jn\omega_0 t} \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

$$\Rightarrow x(t) \cdot e^{-jn\omega_0 t} = \sum_{k=-\infty}^{\infty} c_k e^{-jn\omega_0 t} \cdot e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} c_k e^{j(k-n)\omega_0 t}$$

$$\Rightarrow \int_0^T x(t) \cdot e^{-jn\omega_0 t} dt = \int_0^T \sum_{k=-\infty}^{\infty} c_k e^{j(k-n)\omega_0 t} dt = \sum_{k=-\infty}^{\infty} c_k \int_0^T e^{j(k-n)\omega_0 t} dt$$

$$\Rightarrow \int_0^T x(t) \cdot e^{-jn\omega_0 t} dt = \sum_{k=-\infty}^{\infty} c_k \int_0^T e^{j(k-n) \cdot 2\pi f_0 t} dt$$

$$= \sum_{k=-\infty}^{\infty} c_k \left[\int_0^T \cos(2\pi f_0 \cdot (k-n)t) dt + j \int_0^T \sin(2\pi f_0 \cdot (k-n)t) dt \right]$$

for all values of k except $k=n$, the integrals are 0. Integration over 1 time period

for all sinusoids = 0.

$$\text{So, } \int_0^T x(t) \cdot e^{-jn\omega_0 t} dt = C_n \cdot \int_0^T \cos 0 dt + C_n j \int_0^T \sin 0 dt$$

$$\Rightarrow \int_0^T x(t) \cdot e^{-jn\omega_0 t} dt = C_n T$$

$$\therefore C_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt$$

$$\text{i.e. } c[k] = \frac{1}{T} \int_0^T x(t) \cdot e^{-jk\omega_0 t} dt \longrightarrow \text{contains both amplitude and phase info}$$

$\longrightarrow$ analysis equation

Putting both equations side by side —

$$x(t) = \sum_{k=-\infty}^{\infty} x[k\omega_0] e^{jk\omega_0 t} \longleftarrow \text{time domain (CT)}$$

$$X[k\omega_0] = \frac{1}{T} \int_0^T x(t) \cdot e^{-jk\omega_0 t} dt \longleftarrow \text{frequency domain (DF)}$$

The concept for DT is quite similar. Let's write the naive DT representation of the 2 equations.

$$x[n] = \sum_{k=-\infty}^{\infty} x[k\Omega_0] e^{jk\Omega_0 n}$$

$$X[k\Omega_0] = \frac{1}{N} \sum_{n=0}^{N-1} x[n] \cdot e^{-jk\Omega_0 n} \longleftarrow 0 \text{ to } N-1 \text{ contains } N \text{ points}$$

The analysis formula is perfect. But the synthesis equation has a major flaw.

Remember how DT sinusoids had their frequencies limited between $+\pi$ to $-\pi$ or 0 to $+2\pi$?

This implies $X[k\Omega_0]$ will have repetitions outside of one 2π range. So the synthesis equation with

a k ranging from $-\infty$ to $+\infty$ will greatly overestimate the values because of these repetitions. So we will have to clip the range so that k can take any one unique set of values.

$\therefore$ fundamental frequency $= \Omega_0 = \frac{2\pi}{N} \longrightarrow k\Omega_0$ will be unique from $k=0$ to $N-1$

So the synthesis equation will become —

$$x[n] = \sum_{k=0}^{N-1} x[k\Omega_0] e^{jk\Omega_0 n}$$

→ easier way to get the same thing is by summing or integrating over one time period for periodic signals.

Summarizing —

Time domain		Frequency domain
Continuous + Periodic		Discrete + Aperiodic
$x(t) = \sum_{k=-\infty}^{\infty} x[k\omega_0] e^{jk\omega_0 t}$	$\xleftarrow{\text{CT}} \quad \xrightarrow{\text{FS}}$	$X[k\omega_0] = \frac{1}{T} \int_0^T x(t) \cdot e^{-jk\omega_0 t} dt$
Discrete + Periodic		Discrete + Periodic
$x[n] = \sum_{k=0}^{N-1} x[k\Omega_0] e^{jk\Omega_0 n}$	$\xleftarrow{\text{DT}} \quad \xrightarrow{\text{FS}}$	$X[k\Omega_0] = \frac{1}{N} \sum_{n=0}^{N-1} x[n] \cdot e^{-jk\Omega_0 n}$